पशुवैद्यकीय अधिकारी, महाराष्ट्र पशुवैद्यकीय सेवा, गट-अ, मुख्य परीक्षा
महसूल व वन विभाग
Veterinary Officer, Maharashtra Veterinary Services, Group-A
Main Examination
Revenue and Forest Department

-: परीक्षा योजना :-

१)	लेखी परीक्षा	४०० गुण (एकूण पेपर २)
२)	मुलाखत	५० गुण
एकूण गुण		४५० गुण

विषय	सांकेतांक	गुण	दर्जा	माध्यम	कालावधी	प्रश्नपत्रिकेचे स्वरुप
पेपर-१ पशुसंवर्धन	११८६	२००	पदवी	इंग्रजी	तीन तास	पारंपारिक/वर्णनात्मक
पेपर-२ पशुवैद्यकीय विज्ञान	११८७	२००	पदवी	इंग्रजी	तीन तास	पारंपारिक/वर्णनात्मक

-: अभ्यासक्रम :-

PAPER I : Animal Husbandry

UNIT 1

ANIMAL GENETICS AND BREEDING

Biostatistics and Computer Application: Introduction and importance of statistics and biostatistics, Classification and tabulation of data. Graphical and diagrammatic representation of data. Probability and probability distributions, Moments, Skewness and Kurtosis. Correlation and Regression. Introduction of sampling methods. **Computer Application:** Introduction to computer languages, Data Base Management, Review of MS-Office and its components. Concepts of computer networks, internet & e-mail. **Principles of Animal and Population Genetics:** Mitosis and Meiosis. Chromosome numbers and types in livestock and poultry. Overview of Mendelian principles. Modified Mendelian inheritance. Multiple alleles; lethals; sex-linked, sex limited and sex influenced inheritance. Linkage, crossing over and construction of linkage map. Mutation, Chromosomal aberrations. Cytogenetics, Molecular genetics, Gene concept, DNA and its replication. Introduction to molecular techniques. Genetic structure of population: Gene and genotypic frequency. Hardy-Weinberg law and its application. Quantitative and qualitative genetics; Concept of correlation and interaction between Genotype and Environment. Heritability and Repeatability. Genetic and Phenotypic Correlations. **Principles of Animal Breeding:** Livestock and Poultry Breeding: History Economic characters of livestock and poultry and their importance. Selection, types, response to selection and factors affecting it. Bases of selection. Method of selection, Single and Multi trait. Classification of mating systems. Inbreeding coefficient and coefficient of relationship. Genetic and phenotypic consequences of inbreeding, inbreeding depression, application of inbreeding. Out breeding and its different forms, application of outbreeding and heterosis. Systems of utilization of heterosis. Breeding strategies for the improvement of dairy cattle, buffalo, sheep, goat, swine and poultry. Sire evaluation. Open nucleus breeding system. Current livestock and poultry breeding policies and programmes in the

	state and country. **Methods of conservation-** livestock and poultry conservation programmes in the state and country. Application of reproductive and biotechnological tools for genetic improvement. Breeding of pet, zoo and wild animals Pedigree sheet, selection of breeds and major breed traits. Breeding management of dogs and cats. Common pet birds seen in India and their breeding management. Population dynamics and effective population size of wild animals in captivity or zoo or natural habitats. Planned breeding of wild animals. Controlled breeding and assisted reproduction. Breeding for conservation of wild animals.
UNIT 2	**ANIMAL NUTRITION** **Principles of Animal Nutrition and Feed Technology:** Importance of nutrients in animal production and health. Composition of animal body and plants. Nutritional terms. Nutritional aspect of carbohydrates, protein, fats, vitamins and minerals, their requirements and supplementation in feed. Common feeds and fodders, their classification, availability and importance for livestock and poultry production. Measures of food energy and their applications. Protein evaluation of feeds, biological value of protein. Calorie protein ratio. Nutritive ratio. Introduction to feed technology- Feed industry; Processing of concentrates and roughages. Various physical, chemical and biological methods for improving the nutritive value of inferior quality roughages. Silage and hay preparation. Harmful natural constituents and common adulterants of feeds and fodders. Feed additives in the rations of livestock and poultry. **Applied Ruminant Nutrition:** Importance of scientific feeding. Measurement of digestibility. Feeding standards, their uses and significance, merit and demerits of various feeding standards. Balanced ration and its characteristics. General principles of computation of rations. Formulation of rations and feeding of livestock. **Zoo/wildlife Nutrition:** Feed and feeding of common wild and zoo animals/birds encountered in India. **Applied Non-Ruminant Nutrition:** Nutrient requirements in poultry and other simple stomach animals. Feeding standards for non-ruminants as per BIS and ICAR specifications.
UNIT 3	**VETERINARY PHYSIOLOGY** **Blood, Cardiovascular, Nervous and Muscular Systems:** Properties of blood as a body fluid, Heart- morphological characteristic, systemic excitability conduction. Cardiac Cycle. Electrocardiography and echocardiography- significance in Veterinary Sciences. Haemodynamics of circulation, Major function system- sensory, consciousness, emotion, motor and visceral control. Sense organs and receptors, physiology of special senses. **Digestive and Respiratory Systems:** Morphological characteristic of mono gastric and poly gastric digestive system. Prehension, rumination; defecation; vomition; regulation of secretary function of various organs; hunger, developmental aspects of digestion; luminous, membranous and microbial digestion in rumen and intestine; enzymatic digestion in monogastric and fermentative digestion in rumen. Digestion in birds. Functional morphology of respiratory apparatus. Mechanics of breathing. exchanges of gases in lungs and tissues Respiration in birds. **Excretory and Endocrine Systems:** Kidney- factors determining filtration pressure, methods of studying renal functions; urine concentration; micturition, uremia. Fluid, water balance, fluid therapy, dehydration, water concentration mechanisms. Acid base balance, correction and evolution of imbalances, total osmotic pressure. Formation and excretion of urine of Birds. metabolism of hormones-methods of study of endocrine system. **Reproduction, Lactation, Growth and Environmental Physiology:** Genetic and endocrine control of gonadal development, modification of gonadotrophin release, ovarian functions, follicular development, dynamics, endocrine and receptor profiles, sexual receptivity, ovarian cycle, post-partum ovarian activity, ovum transport, hormones present in the biological fluids during pregnancy and their uses for the diagnosis of pregnancy- maternal foetal placental participation in pregnancy and parturition, immunology of gestation, preparturient endocrine status. Spermatogenic cycle and wave- function of sertoli cell-leydig cell, cryptorchidism. Puberty - photoperiod - uses of androgens, progestogens, estrogens.

	Functional and metabolic organization of mammary glands; lactogenesis and galactogenesis; prolactin and lactation cycle. Heat balance, heat tolerance, thermo-regulation in farm animals, role of skin, responses of animals to heat and cold, fever, body temperature and hibernation. Temperature regulation in birds.
UNIT 4	**LIVESTOCK PRODUCTION MANAGEMENT** **General Livestock Management:** Problems and prospects of livestock industry in India. Common animal husbandry terms. Transportation of livestock and wild or zoo animals. Common farm management practices. Culling of animals. Selection and purchase of livestock. **Fodder Production and Conservation:** Importance of grasslands and fodder in livestock production. Soil and Water conservation and drainage of water for fodder production. Fodder production for livestock. Storage of feeds and fodders. Scarcity fodders and preservation of green fodder. **Livestock Production Management-Ruminants:** Housing systems, General principles affecting the design and construction of building for housing for various livestock species. Breeds and economic traits of cattle and buffalo and descriptions of important breeds. General management and feeding practices of animals, bulls and working animals. Draught ability. Routine animal farm operations and labour management, farm accounts and records. Milking Methods. Factors affecting quality and quantity of milk production. Breeds of sheep and goat and their descriptions. **Zoo Animals Production Management:** Taxonomy of important wild animals, reptiles, amphibians and birds, Status and conservation practices of wild life in India. Different methods of Wildlife Census. Basic principles of habitat and housing of various classes of wild zoo animals. Size and space requirement of cubicles, enclosures of important wild zoo animals. Management of livestock in fringe areas, in and surrounding the breeding areas. Feeding habits, feeds and feeding schedules of captive wild animals. **Restraint, Rescue, organizations in wildlife management:** Restraining, capture, handling, physical examination of captive animals. Man-wildlife conflict and its mitigation, Rescue and rehabilitation of wildlife, Rapid Response Team (RRT), Functions of Transit Treatment Centre (TTC). Classification of zoos, management of sanctuaries, national parks etc. Acts and Rules related to captive animals. National and international organization and institutions interlinked to captive animals/wildlife conservation- Wildlife Institute of India (WII), Wildlife Preservation Society of India, Central Zoo Authority, World Wide Fund for Nature- India, Wildlife Trust of India, Wildlife Research & Training Centre, MAFSU, Nagpur- role and functioning. Wildlife Protection Act 1972. **Animal Welfare:** Definition of animal welfare and ethics. Welfare in relation to ecosystem and environmental factors. Role of veterinarians in animal welfare. Animal welfare organizations, Animal Welfare Board of India - their role, functions and current status. Rules, regulations, laws on animal welfare. Prevention of Cruelty to Animals Act, 1960 (59 of 1960). Role and function of Committee for Control and Supervision of Experiments on Animals. Protection of wild life in nature and captivity. Welfare of animals during transportation. Protection and welfare of working animals, pets and companion animals. Animal welfare during natural calamities and disaster management. Legal duties of veterinarians, Common offences against animals and laws related to them. Provincial and Central Acts relating to animals. Laws relating to offences affecting Public Health. Livestock Importation Act. Evidence, liability and insurance. Code of Conduct and Ethics for veterinarians - the Regulations made under the Act. **Poultry Production Management:** Indian poultry industry – Brief outline. Classification of poultry with respect to production characters, age and standards. Production characters of other avian species. Description of indigenous fowls and their value in rural farming. Brooding management – Types of brooders – preparation of shed – Importance of environmental factors. Housing – Types of poultry houses – space requirements. Scavenging system of management – Low input technology – Backyard and semi-intensive units; their management and economic achievements. Deep litter management. Cage management– Different types, Advantages and disadvantages. Management of growers, layers, broilers and breeders. Management of stress, feeding and Watering. Breeding systems and methods of mating. Poultry judging. Egg structure. Bio-

	security and principles of disease prevention management. **Diversified Poultry Production and Hatchery Management:** Principles of incubation and hatchery management practices. Factors affecting fertility and hatchability, hatchery hygiene. Candling, sexing, grading, packing of eggs and disposal of hatchery waste. Organic and hill farming. Mixed or integrated poultry farming. Management of ducks, geese, turkeys, Japanese quails, and guinea fowls. **Laboratory or Pet Animal Production Management:** Importance and selection of laboratory animal, care and housing standards of mice, rats, hamster and guinea pigs. General considerations on feeding and breeding. Scope of rabbit farming in the country. Important breeds of dogs, cats and pet birds and their feeding. **Swine and Equine Production Management:** Introduction and scope of swine farming in the country. Important breeds (exotic and indigenous) & their characteristics. Housing and feeding of swine for optimal production. Equine population of India. Horses, donkeys and mules and their utility. Dentition and ageing of horses. Care and routine management of equines including grooming, saddling and exercise. Stable and its management. Vices of horses. Foot care and shoeing care. Feeding routine for equines. Care of stallion. Mating of horses, broodmare and its care. Foaling and care of newborn. Breeding mules. Doping and its detection. Colic and its prevention.
UNIT 5	**LIVESTOCK PRODUCTS TECHNOLOGY** **Milk and Milk Products Technology:** Layout of milk processing plant and its management. Physico-chemical properties of milk. Microbiological deterioration of milk and milk products. Collection, chilling, standardization, pasteurization, UHT treatment, homogenization, and bactofugation. Dried, dehydrated and fermented milk. Common defects of milk products. Packaging, transportation, storage and distribution of milk and milk products. Organic milk products. Food safety standards for milk and milk products. Dairy effluent management. **Wool Science:** Introduction to wool, fur, pelt and specialty fibers with respect to processing industry. Glossary of terms of wool processing. Post-shearing operations of wool, classification and grading of wool, physical and chemical properties of wool. Impurity of wool, factors influencing the quality of wool. **Abattoir Practices and Animal Byproducts Technology:** Layout and management of rural, urban and modern abattoirs. HACCP concepts in abattoir management. FSSA standards on organization and layout of abattoirs. Animal welfare and pre-slaughter care, handling and transport of meat animals including poultry. Slaughtering and dressing of meat animals and birds. Evaluation, grading and fabrication of dressed carcasses including poultry. Abattoir byproducts; rendering, meat, bone, glue, gelatin, fat and byproducts of pharmaceutical value. Skin and hides; methods of flaying, defects, preservation and tanning. Treatment of condemned meat and carcasses. Management of effluent emanating from abattoir. **Meat Science:** Prospect of meat industry in India. Fraudulent substitution of meat. Preservation of meat and poultry; drying, salting, curing, smoking, chilling, freezing, canning, irradiation and chemicals. Ageing of meat. Modern processing technologies of meat and meat products. Packaging of meat and meat products. Formulation and development of meat sausages, meat balls or patties, tandoori chicken, soup, pickles. Fermentation of meat products. Basics of sensory evaluation of meat products. Nutritive value, preservation, packaging of egg and egg products. Laws governing national or international trade in meat and meat products.
UNIT 6	**VETERINARY AND ANIMAL HUSBANDRY EXTENSION EDUCATION** **Livestock based Livelihoods and their Evolution:** Evolution and relationship between agriculture and animal husbandry. Farming and characteristics of farming in India. Classification of farming, types and systems. Peasant farming, cooperative farming, collective farming, contract farming, estate farming, organic farming, capitalistic farming, small-scale farming, large-scale farming, intensive, extensive farming, specialized, diversified, mixed, integrated and dry land farming. **Extension Education and Development:** Types of education: Formal, non-formal and informal education. Extension education: Concept, levels, objectives and dimensions. Teaching- learning process and steps in extension teaching. Rural development - Concept, significance and importance of

	rural development programmes. Problems and Issues in development. Panchayati Raj System. **Rural Sociology in Veterinary Extension:** Concept of sociology and rural sociology in animal husbandry extension. Social institutions in rural society: Social, economic, political, religious and educational (definition, composition and function). Social change: concept, importance and factors. Leadership: definition, functions of leader, types of rural leaders. **Transfer of Technology for Livestock Development:** Technology- Concept, generation process, application, merits and de-merits. Adoption and diffusion of innovations, stages of adoption, adopter categories, innovation decision process, attributes of innovations, diffusion process, factors affecting adoption and diffusion processes. Programme planning- principles and objectives. Constraints in the adoption of scientific animal husbandry practices. Cattle and buffalo improvement programmes: Key Village Scheme, Intensive Cattle Development Project, Gosadan and Gaushala. Dairy development programmes: concept of cooperation, Amul pattern of dairy cooperative system and Operation Flood. Transfer of technology projects of Indian Council of Agricultural Research: Krishi Vigyan Kendra, Agricultural Technology Information Centre, Agricultural Technology Management Agency, National Agricultural Innovation Project (NAIP), Rashtriya Krishi Vikas Yojana etc. Different ongoing central and state government animal husbandry development programmes being run related to livestock. **Communication and Extension Teaching Methods:** Communication and its functions, . Types of communication. Individual contact methods, group contact methods and Mass contact methods. Selection and use of extension teaching methods. **Livestock Economics and Marketing:** Economics and Livestock Economics: definition and scope. Basic concepts- wants, goods, wealth, utility, price, value, assets, capital, money, income etc. Theories of demand, supply and cost. Theories of production. Concept of market. Government interventions and role in marketing of livestock and livestock products. External trade in livestock products, recent policies on trade and international trade agreements and their implications in livestock sector. **Livestock Entrepreneurship:** Definition of and difference between entrepreneur, entrepreneurship, enterprise and manager. Theories of entrepreneurship, characteristics and functions of an entrepreneur. Forms of entrepreneurship. Project appraisal-Introduction, importance, techno-economic feasibility, criteria of project evaluation, capital budgeting, etc. Business plan for enterprise. Institutions promoting entrepreneurship in India. Entrepreneurship development programmes. **Information and Communication Technology:** Strengths and limitations of ICTs application in livestock sector and farmers capacity building. Information kiosk, E- learning, CAD, virtual class room, virtual reality, multi-media etc. Cyber extension- problems and prospects in livestock extension. **Contemporary Issues in Livestock Enterprises:** Role of women in animal husbandry, gender sensitization and its importance, need for gender analysis, gender budgeting and mainstreaming. Livestock census, insurance scheme, national livestock mission.

PAPER II : Veterinary Science	
UNIT 1	**VETERINARY ANATOMY** Introduction to anatomy and branches of anatomy and descriptive terms used in anatomy and study of anatomical planes. **General Osteology, Arthrology and Myology:** Study of properties, classification and structure of bone. Introduction to arthrology, classification of joints, different diarthrodial joints. Introduction to myology, classification of muscles. **General Angiology, Neurology and Aesthesiology:** Introduction to angiology. General plan of systemic and pulmonary circulations, lymphatic and venous systems. Introduction to neurology and parts of central, peripheral and autonomic nervous system and sense organs. **General Splanchnology:** Boundaries of thoracic, abdominal and pelvic cavities, topography of different organs of digestive, respiratory, urinary, endocrine, male and female reproductive systems of domestic animals and fowl. **Fore limb:** Study of bones of fore limb of ox and differences in horse, dog, pig and fowl. **Head and neck:** Study of cranial and facial bones, cervical vertebrae of ox and differences in horse, dog, pig and fowl. Muscles of face. Study of teeth, hard and soft palate, tongue, pharynx, larynx, thyroid, parathyroid and salivary glands. Study of sense organs, trachea and oesophagus. **Thorax:** Study of thoracic vertebrae, ribs and sternum of trachea, thymus, oesophagus, lungs ox and differences in horse, dog, pig and fowl. **Abdomen:** Study of bones of abdomen of ox and differences in horse, dog, pig and fowl. Blood and nerve supply to abdominal viscera. Study of organs of digestive, urinary, male and female reproductive systems present in abdomen and differences in horse, dog, pig and fowl. Study of mammary glands and spleen. **Hind limb and pelvis:** Study of bones of hind limb and pelvis of ox and differences in horse, dog, pig and fowl. Study of joints, ligaments, blood vessels, lymph nodes and nerves of hind limb, pelvis and tail region and pelvic viscera. Study of external genital organs. **Embryology:** Introduction to embryology, gametogenesis, fertilization, cleavage, types of eggs, morula, blastulation, gastrulation, types of implantation.
UNIT 2	**VETERINARY GYNAECOLOGY AND OBSTETRICS** **Veterinary Gynaecology:** Bovine, ovine, caprine, and feline Equine : Puberty and sexual maturity and their endocrine control- Delayed puberty- Its causes, clinical approach, treatment and prevention of delayed puberty, Oestrous cycle and factors affecting length of the oestrous cycle, Aberrations of oestrus and their clinical management and problems in oestrus detection, Follicular Dynamics and its clinical impact on fertility improvement, Incidence causes, diagnosis, treatment and prevention of ovulatory failures, Fertilization and aberrations of fertilization. Pathological affections of female genitalia, various affections of genital organs and their medicinal and surgical management. Assisted reproductive techniques: Synchronization of estrus and ovulation, Multiple ovulation and Embryo transfer technology, In-vitro fertilization. Methods of Population control by medical and surgical techniques. Principle, procedure and application of ultrasonography in farm and pet animal reproduction.**Veterinary Obstetrics:** Farm and pet animals - Maternal recognition of pregnancy, Pregnancy diagnosis, Duration of pregnancy, Factors affecting gestation length, Care and management of pregnant animals, Implantation, Placentation, Classification, functions, Superfetation and Superfecundation, Clinical management of specific and nonspecific causes of abortion and various conditions affecting genital organs. Parturition, signs and stages, Initiation and induction of parturition, Puerparium and factors affecting puerparium, Post-partum care of the dam and neonate in different species of farm and pet animals, Dystocia: Classification, Clinical signs, diagnosis and treatment of dystocia in small and large animals. Post-partum uterine infections, Post-partum resumption of ovarian activity .**Veterinary Andrology & A.I.** Farm and pet animals - Comparative clinical reproductive anatomy and endocrinology of the male reproduction, Common congenital and genetic defects of the male reproductive tract, Puberty and sexual maturity and factors affecting them, Sexual behaviour and libido, Sperm transport, erection and ejaculation, Coital injuries and vices in male

	animals, Semen collection techniques, Structure of Spermatozoa, Semen evaluation, Semen extenders, dilution, preservation and post thaw evaluation, Artificial insemination techniques in farm and pet animals, Forms of male infertility. Affections of the scrotum, testis, accessory sex glands, penis and prepuce, Breeding soundness evaluation of bull, *In vitro* tests for evaluation of male fertility, Medical and surgical techniques for population control of the male reproduction.
UNIT 3	**VETERINARY MEDICINE** **General:** Concepts of diagnosis, differential diagnosis, treatment and prognosis. General systemic states, hyperthermia, hypothermia, fever, septicemia, toxemia, shock, allergy, anaphylaxis, oedema, coma, anaemia, common clinical poisonings and dehydration. **Systemic Diseases:** Etiology, clinical manifestations, diagnosis, differential diagnosis, treatment, prevention and control of the various diseases of cattle, buffalo, sheep, goat, horse, pig, dog, cat and poultry. Emergency medicine and critical care. **Metabolic and Deficiency Diseases:** Diagnosis and management of diseases caused by deficiency of macro and micro minerals, vitamins and management. Alternative or integrated or ethno veterinary medicine in animal disease management. Aetiology, clinical manifestations, diagnosis, differential diagnosis, treatment prevention and control of metabolic or production and endocrine diseases of livestock, companion animals and poultry. **Zoo and Wild Animal Medicine:** Principles of wildlife health management, Principles of zoo hygiene, role of wildlife veterinarian in wildlife conservation, public health problems arising from zoos. Prevention, control and treatment of bacterial, viral, parasitic, nutritional and metabolic diseases in zoo and wild animals including exotic birds. **Bacterial, Fungal and Rickettsial Diseases:** Aetiology, epidemiology, clinical manifestations, diagnosis, treatment, prevention and control of bacterial, fungal and rickettsial diseases of livestock and poultry. **Viral and Parasitic Diseases:** Aetiology, epidemiology, clinical manifestations, diagnosis, treatment, prevention and control of viral and parasitic diseases of diseases of cattle, buffalo, sheep, goat, horse, pig, dog, cat and poultry. Parasitic diseases and treatment of trematodes, cestodes, nematodes, protozoan infections and external parasites of clinical importance. **Jurisprudence, Ethics and Animal Welfare:** Legal duties of veterinarians, laws related to medicine, evidence, common offences against animals and laws related to these offences. Examination of living and dead animals in criminal cases. Cruelty to animals and bestiality. Legal aspects of: Examination of animals for soundness, examination of injuries and post-mortem examination. Collection and dispatch of materials for chemical examination, detection of frauds-doping, alternation of description, bishoping etc. Cattle slaughter and evidence procedure in courts. Provincial and Central Acts relating to animals. Glanders and Farcy Act 1899 (13 of 1899). Dourine Act 1910 (5 of 1910). Laws relating to poisons and adulteration of drugs. Livestock importation act, liability and insurance. Code of conduct and ethics for veterinarians-the regulations made under the Act.
UNIT 4	**VETERINARY PARASITOLOGY** **General Veterinary Parasitology:** Introduction, Types of parasites and hosts and vectors. Types of animal associations. Modes of transmission of parasites and methods of dissemination of the infective stages of the parasites. Immunity against parasitic infections or infestations. General harmful effects of parasites including various tissue reactions caused by parasites. General control measures against parasites. **Trematodes, Cestodes, nematodes and Arthropods.** Introduction, general account, general life cycle with morphological features of their developmental stages. Important morphological features, lifecycle, modes of transmission, pathogenesis, epidemiology, diagnosis and general control measures of Trematodes, Cestodes, nematodes and Arthropods. **Protozoa:** Introduction, general account and classification, general life cycle of protozoa with morphological features of their developmental stages. Important morphological features, life cycles, modes of transmission, pathogenesis, epidemiology, diagnosis and general control measures of the protozoan parasites of veterinary and zoonotic importance.

UNIT 5	**VETERINARY PHARMACOLOGY AND TOXICOLOGY** **General Pharmacology:** Pharmacological terms and definitions, nomenclature of drugs. Principles of drug activity: Pharmacokinetics - Routes of drug administration, absorption, distribution, biotransformation and excretion of drugs. Pharmacodynamics- Concept of drug and receptor, dose-response relationship, terms related to drug activity and factors modifying the drug effect and dosage. Adverse drug reactions, drug interactions. Drugs acting on Autonomic Nervous System, Central Nervous System and different body systems. **Veterinary Chemotherapy**: Antimicrobial agents: Classification, general principles in antimicrobial chemotherapy, antimicrobial resistance, combined antimicrobial therapy. Pharmacology of indigenous medicinal plants: Scientific name, common name, active principles and pharmacological actions and therapeutic uses. **Veterinary Toxicology:** General Toxicology: Definitions in toxicology. Sources and classification of toxicants, factors modifying toxicity, general approaches to diagnosis and treatment of poisoning. Toxicity caused by metals and non-metals, poisonous plants, agrochemicals, fungal and bacterial toxins. Venomous bites and stings. Toxicity caused by food additives and preservatives. Drug and pesticide residue toxicology. Environmental pollutants. Concept of radiation hazards.
UNIT 6	**VETERINARY PATHOLOGY** **General Veterinary Pathology:** Introduction and scope of Veterinary Pathology. Major intrinsic and extrinsic causes of disease. Haemodynamic disorders. Reversible and irreversible cell injury, differences between post-mortem autolysis and necrosis, gangrene and its types. Photosensitization. Disturbances in growth. Inflammation: Definitions, classification, various cell types and their functions, mediators, cardinal signs and systemic effects. Wound healing by primary and secondary intention including growth factors. Immunodeficiency, hypersensitivity and autoimmunity. **Systemic Veterinary Pathology:** Pathological changes affecting Digestive, Respiratory, Musculoskeletal, Cardiovascular, Haematopoietic, Lymphoid, Urinary, Reproductive, Nervous, Endocrine systems, Skin and Appendages, Ear and Eye. **Animal Oncology, Veterinary Clinical Pathology and Necropsy:** Definitions, general characteristics and classification of neoplasms. Differences between benign and malignant tumours, aetiology, carcinogenesis and spread of neoplasms, tumour immunity, effects and diagnosis of tumours, staging and grading of neoplasms. Pathology of various types of tumours in domestic animals. **Veterinary Clinical Pathology:** Haematology – Different anticoagulant used in haematology, interpretation of blood tests, blood smear examination and its interpretation. Urinalysis- Interpretation of physical, chemical and microscopic examination of urine. Study of biopsy and cytology as rapid diagnostic techniques. **Necropsy:** Introduction, objectives, pre-necropsy guidelines, procedure for post-mortem examination of various species of animals including wild animals, post-mortem changes, collection, preservation and dispatch of specimens (morbid materials) for laboratory examination, writing of post-mortem report, veterolegal necropsy. **Pathology of Infectious and Non-Infectious Diseases of Domestic Animals, laboratory animals, wild animals and Avians:** Pathogenesis, gross and microscopic pathology of important pathologic bacterial, viral, fungal, protozoan and prion diseases of livestock and poultry. Pathogenesis, gross and microscopic pathology of nematodes, cystodes and trematodes, Pathological changes of nutritional imbalances due to carbohydrates, proteins, fats, minerals and vitamins and metabolic diseases. Gross and microscopic pathology of toxicities like arsenic, copper, lead, mercury, cadmium, strychnine, nitrate or nitrite, hydrocyanic acid, fluoride, selenium and oxalates; insecticide or pesticide poisoning, plant poisoning.
UNIT 7	**VETERINARY MICROBIOLOGY** **General & Systematic Veterinary Bacteriology:** Classification, nomenclature and structure of bacteria; Bacterial stains and techniques; Growth and nutritional requirement of aerobic and anaerobic bacteria; Types and sources of infection, method of transmission

	of infection. Pathogenicity, virulence, Epizootic and enzootic diseases, bacteremia, septicaemia and toxaemia, endotoxins, exotoxins, antitoxins, toxoids; Bacterial genetics, plasmids and antibiotic resistance. Characteristics of the important pathogenic bacterias in relation to isolation, growth, cultural, morphological, biochemical and antigenic characteristics. **Veterinary Mycology:** Introduction, classification, general properties of fungi; Growth and Reproduction of fungi; Characteristic of important pathogenic fungi, epidemiology, pathogenesis, diagnosis and control of fungal diseases. **Microbial Biotechnology:** Basic concepts and scope of Recombinant DNA technology; Gene cloning, Cloning vectors and expression vectors; Transformation and transfection; Southern, Northern and Western blotting; Bioinformatics, Gene banks; Application of molecular and biotechnological techniques, IPR. Ethics and regulatory issues in Animal Biotechnology. **Veterinary Immunology and Serology:** History of Immunology; Lymphoid organs, tissues and Cells: Types of Immunity; Antigens, blood group Antigens, Mitogens and factors affecting immunogenicity; Adjuvants; Antibody: Structure, physiochemical properties and functions of various classes of immunoglobulins, Hybridoma and monoclonal antibodies, Serological reactions. Major histocompatibility complex (MHC) structure, function and gene organization; Cytokines. **General and Systematic Veterinary Virology:** Introduction to viruses; Structure and Classification of Viruses; Replication and pathogenesis, Oncogenesis, latency and immunopathology. Studies on General Properties, Antigens, Cultivation and pathogenesis, emerging, re-emerging and transboundry viruses and Viral Infections.
UNIT 8	**VETERINARY SURGERY AND RADIOLOGY** **Veterinary General Surgery:** Classification of surgery, tenets of Halsted. Pre-operative, intra- operative and post-operative considerations. Sterilization and disinfection, Sutures and suture materials, Treatment of acute and chronic inflammation, Hemostasis, Basic surgical affections and treatment of abscess, tumour, cyst, hernia, hematoma, necrosis, gangrene, burn and scald, frost bite, sinus and fistula, wounds, Introduction to biomaterials and stem cell therapy in wound management. Management of surgical shock. Principles of fluid therapy in surgical patients. **Veterinary Anaesthesiology:** Terminology, classification and indications in anaesthesiology. General considerations of anaesthesia, Local and regional anaesthesia, Premedication, properties and use. Uses of premedication, anticholinergic, sedatives and tranquilizers. General anaesthesia-Intravenous and inhalation and dissociative anaesthesia, monitoring of anaesthesia. Avian, wild, zoo, exotics and lab animal anaesthesia and capture myopathy. Anaesthetic emergencies and management, Toxicity, antidote and reversal agents. **Wild or zoo animal restraint**: Psychological restraint, physical and chemical restraint. Considerations for wildlife anaesthesia, various physical capture devices, darting equipments, their advantages and disadvantages, different drugs used in chemical restraints, their doses and reversal.**Veterinary Diagnostic Imaging Techniques:** Introduction to Radiology-General terminology of radiology, Production of X-rays and factors influencing production of X-rays. Radiation hazards and safety measures. Contrast radiography. Diagnostic ultrasonography. Advanced diagnostic imaging tools- CT scanning and MRI. **Regional Surgery I:** Affections of Head and Neck: general considerations, etiology, pathogenesis, anaesthetic protocol, surgical procedure and post-operative care of different affections , **Regional Surgery II:** Thorax and Abdomen- general considerations, etiology, pathogenesis, anaesthetic protocol, surgical procedure and post-operative care of different affections. **Orthopedics and Lameness:** Body conformation of the horse in relation to lameness, various terms used, (forelimb and hind limb). Various affections of fore limb and hind limb of livestock and companion animals (fracture/dislocation and other conditions)- etiology, diagnostic methods, conservative and surgical management along with preventive measures. Intervertebral disc disease in dogs. Rehabilitation and physiotherapy of orthopaedic patients.
Unit 9	**VETERINARY PUBLIC HEALTH AND EPIDEMIOLOGY** **Veterinary Public Health and Food Safety** Role of veterinarians in public health. One

	Health concept and initiatives. Sources of contamination. Principles and concepts of food hygiene and safety. Milk and meat hygiene in relation to public health, safe milk and meat production, adulteration and residual toxicants, Milk and meat plant and equipment hygiene. Quality control of products of animal origin. Milk and meat hygiene practices in India and other countries. Elements of meat inspection and meat hygiene practices. Pathological conditions associated with the transport of food animals. Hygiene in abattoirs and meat plants. Detection of conditions or diseases and judgements during ante-mortem and post-mortem inspection. Meat as a source of disease transmission. Classification of low risk and high risk material generated in an abattoir and its hygienic disposal. Inspection of poultry for human consumption. Occupational health hazards in abattoir and meat plants. Food-borne infections and intoxications associated with foods of animal origin. Types of biohazards. Hazard analysis and critical control points (HACCP) system. Risk analysis, assessment and management. International food safety standards, Sanitary and phytosanitary measures in relation to foods of animal origin. Food Safety and Standards Act and Regulations. Role of Food Safety and Standards Authority of India (FSSAI), Bureau of Indian Standards (BIS) and other national agencies. **Veterinary Epidemiology:** Definitions, components and aims of epidemiology. Factors influencing occurrence of livestock diseases and animal production. Measures and patterns of disease occurrence. Survey and surveillance of animal diseases and related parameters. Epidemiological methods. Animal disease forecasting. Strategies of disease management: prevention, control and biosecurity. Economics of animal diseases. National and international regulations on livestock diseases. Role of OIE and laws on international trade of animals and animal products. **Zoonotic Diseases:** Definition of zoonotic diseases. Classification of zoonoses and approaches to their management. Multi-sectoral approach for zoonoses prevention and control. Emerging, re-emerging and occupational zoonoses. Role of domestic, wild, pet and laboratory animals and birds in transmission of zoonoses and different diseases of zoonotic importance in livestock, companion animals and poultry..**Environmental Hygiene:** Scope and importance. ecosystem, biodiversity, natural resources, environmental pollution: nature of pollutants and effects on animal and human health. Air pollution in animal houses, effect on health and productivity. Airborne and waterborne diseases – Classification, health hazard, prevention and control. Soil, marine and thermal pollution- Classification, sources, hazard, prevention and control. Nuclear hazards or radiological hazard radiation protection. National rules and legislations related to environmental pollution and role of Pollution Control Board in India.
UNIT 10	**VETERINARY BIOCHEMISTRY** **General Veterinary Biochemistry:** Scope and Importance of Biochemistry. Biological Significance of Important Monosaccharides, Disaccharides, Polysaccharides, and Mucopolysaccharides, Properties and biological significance of simple, compound and derived lipids and lipoproteins, Classification, Structure, Properties - Biological significance of proteins. Physical and chemical properties of amino acids. Structures and functions of deoxyribonucleic acid (DNA) and a typical ribonucleic acid (RNA). **Intermediary Metabolism:** Enzymes: Definition and classification. Coenzymes, cofactors and iso-enzymes. Carbohydrate metabolism: Glycolysis, Kreb's cycle, HMP shunt, gluconeogenesis, glycogenesis, glycogenolysis, Lipid metabolism: Beta oxidation of fatty acids, ketone body formation, biosynthesis of fatty acids.Protein metabolism: Biosynthesis and Degradation. Ammonia transport and urea cycle. Nucleic acid metabolism: Metabolism of purines and pyrimidines. DNA and RNA biosynthesis and regulation. Regulation and Integration of metabolism. **Veterinary Analytical Biochemistry:** Disorders of Carbohydrate Metabolism, Hormonal control of carbohydrate metabolism and regulation of blood sugar. Biochemical tests for the detection of disturbance in carbohydrate metabolism. Plasma Proteins and clinical significance, Lipid Profile in disease diagnosis. Diagnostic importance of non-functional plasma enzymes and Isoenzymes, Liver function tests, Biochemical tests for differential diagnosis. Biochemical tests of renal function, Urine analysis. Disturbance in acid base balance and its diagnosis. Biochemistry of digestive disorders. Biochemical basis of fluid therapy.